February 8, 2008

Biochemistry Cume
Enzymes

1. Transform the Michaelis-Menten equation into a linear plot of "**v**" *versus* **v/ [S]** and determine the slope and the y-intercept of the plot (**20 points**)

2. An enzyme has a mass of 240 kDa and consists of six identical subunits. For this enzyme, a unit of activity is defined as the amount of enzyme that hydrolyzes 100 μmoles of the substrate in 20 minutes at 37 °C. The purified enzyme has a V_{max} of 2000 units per milligram of enzyme. **25 (10 + 10 + 5) points**

   a. How many moles of substrate are hydrolyzed per second per milligram of enzyme when substrate concentration is much greater than K_M?
   b. How many moles of active sites are there in 10 mg of enzyme? Assume that each subunit has one active site?
   c. Calculate the turnover number of the enzyme?

3. **A**. Define the Michaelis-Menten constant. Given a kinetic scheme,

   $E + [S] \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} [ES] \underset{k_{-2}}{\overset{k_{+2}}{\rightleftharpoons}} E + [S]$, Show that the Michaelis-Menten constant (K_M) is, $(k_{-1} + k_{+2})/k_{+1}$.

   B. An enzyme was assayed at an initial substrate concentration of 15×10^{-5} M. Half the substrate was consumed in 6 minutes. The K_M of the enzyme for the substrate is 2×10^{-5} M. Calculate the, **A**. the rate constant of the reaction; **B**. The V_{max}, of the reaction, and **C**. The concentration of the product formed after 10 minutes of initiation of the reaction.. **30 (15 + 15) points.**

4. Explain why a graph of "**v**" *versus* **[S]** for a typical unisubstrate enzyme catalyzed reaction.cannot provide accurate estimates of the K_M and V_{max} values. Draw "**v**" *versus* **[S]** curves representing an enzyme catalyzed reaction, **a** – in the absence of inhibitor; **b** – in the presence of a competitive inhibitor; **c** – in the presence of a non-competitive inhibitor. (**25 points**)